

Labyrinth

Time limit: 1.00 s

Memory limit: 512 MB

You are given a map of a labyrinth, and your task is to find a path from start to end. You can walk left, right, up and down.

Input

The first input line has two integers n and m : the height and width of the map.

Then there are n lines of m characters describing the labyrinth. Each character is . (floor), # (wall), A (start), or B (end). There is exactly one A and one B in the input.

Output

First print "YES", if there is a path, and "NO" otherwise.

If there is a path, print the length of the shortest such path and its description as a string consisting of characters L (left), R (right), U (up), and D (down). You can print any valid solution.

Constraints

- $1 \leq n, m \leq 1000$

Example

Input	Output
5 8 ##### #.A#...# #.#.#B# #.....# #####	YES 9 LDDRRRRRU